

Course Basics

Course: CMP_SCI 1050 and CMP_SCI 1050H - Algorithm Design and Programming I

Meeting Location: MEMORIAL STUDENT UNION S107

Meeting Time: Tuesdays and Thursdays 9:30 AM – 10:45 AM

Instructor

Name	Dr. Ekinan (Ekin) Ufuktepe
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Teaching Assistants (Check Canvas for Office Hours times, locations)

Name
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Labs (ARTS & SCIENCE 110 (ALLEN AUD) In-Person)

Lab	Days & Times
Regular	Fridays 3:00 PM – 5:30 PM
Honors	Fridays 3:00 PM – 5:30 PM

Honors Philosophy and Goals – for CS 1050H

Students are most engaged and learn the most when they are challenged. As a beginning course, it is not uncommon for honors students to find this course to be rudimentary. The goal of the honors section is to provide advanced students the opportunity to work on more stimulating assignments, with the hope that these students are challenged and find the assignments interesting. The student who asks for more is the student who excels in future courses. Note that the honors lab section will be separate from the non-honors sections and will include dedicated TA/PLA resources to answer questions.

Prerequisites**C- or higher in Math 1100 or 1160 or 1500**

CS1050 requires the ability to think abstractly, including the concepts of functions, variables, and indirection. A good background equivalent to success in College Algebra should mean that a student is prepared to take this course.

Required Textbook

C How to Program (9th Edition), Deitel and Deitel, ISBN 9780137454372

<https://deitel.com/c-how-to-program-9-e/>

<https://www.amazon.com/How-Program-2-downloads-Paul-Deitel-ebook/dp/B09245HVQ5>

<https://www.vitalsource.com/referral?term=9780137398454>

AutoAccess

Mizzou Digital Content Solution

This course is part of our AutoAccess program designed to reduce the cost of course materials for students. You will be able to access the digital content for this course through Canvas on the first day of class automatically.

Your student account will be charged for the cost of the digital course material. We have helped save students over \$67 million by providing digital content over the last 11 years.

The lowest cost content has been sourced. If you choose to opt out of the content, please do so by **Sept 9, 2025** to receive a refund. You will be sent a Welcome Email that will provide charge amounts, the opt-out process and any additional information needed for your AutoAccess course(s) beginning August 11, 2025.

Your AutoAccess course may have a Print Upgrade available as an additional purchase. This is a low-cost version of the printed text made available by the publisher at a reduced cost. It is the publisher's requirement that in order to purchase the additional print, you must be opted in for the AutoAccess digital required material. If you have questions about Print Upgrades and opting out, please contact us at autoaccess.themizzoustore.com.

If you have questions please call 573-882-7611, email AutoAccess@umsystem.edu or visit autoaccess.themizzoustore.com.



Optional Textbook

C Programming Language (2nd Edition), Kernighan and Ritchie, ISBN 9780131103627

<https://www.amazon.com/Programming-Language-2nd-Brian-Kernighan/dp/0131103628/>

Course Details**Codes from the lectures:** https://github.com/ekincanufuktepe/CMP_SC1050**Topics**

1. Programming, compiling and executing with C
2. Algorithms and Control Structures
3. Structured Programming, Testing procedures in programming
4. Functions in C
5. Branching/ Conditional statements: if, if-else, switch, ...
6. Iteration/ Loops: while, for, ...
7. Pointers in C
8. Arrays
9. Strings/String manipulation
10. File I/O, Memory Allocation
11. Structures

Departmental Curriculum Guidelines: Course Learning Objectives

1. Students will have a working knowledge of simple algorithms with flow control, loops, recursion, modular methods and user interaction.
2. Students will show a working knowledge of basic data structures, operations and language concepts including one- and two-dimensional arrays, string manipulation, and set operations.
3. Students will achieve proficiency in the C programming language, including syntax, semantics, and show a working knowledge of the usage of pointers and basic memory management concepts (allocation, use, storage, freeing).
4. Students will be able to incorporate input/output handling to a terminal display, streams or sequential files and exception handling as part of working programs to understand the importance of effective interaction between a program and the user.
5. Students will learn how to build (including make files), debug, profile, test, and manage programs.
6. Students will learn to use professional coding practices and documentation methods.
7. Students will learn testing strategies for program validation, bounds checking and input validation.

Self Determination as a Learner

Learning is the act of acquiring new, or modifying and reinforcing, existing knowledge, behaviors, skills, values, or preferences and may involve synthesizing different types of information. The ability to learn is possessed by humans, animals and some machines. Progress over time tends to follow learning curves. Learning is not compulsory; it is contextual. It does not happen all at once, but builds upon and is shaped by what we already know. **To that end, learning may be viewed as a process, rather than a collection of factual and procedural knowledge.** Learning produces changes in the organism and the changes produced are relatively permanent.

Psychology, 2nd edition. pg. 264.

By Schacter, Gilbert, and Wegner (2009, 2011). Worth Publishers.

Grading and Coursework

The University of Missouri uses an A-F and plus/minus grading system. Letter grades calculated in the grade point average are:

A+ (4.00)	B (3.00)	C- (1.7)	F (0)
A (4.00)	B- (2.7)	D+ (1.3)	
A- (3.7)	C+ (2.3)	D (1.00)	
B+ (3.3)	C (2.00)	D- (0.7)	

Categories and weights

This course uses a default grading scheme, but also offers students the opportunity to adjust grade weights within specified limits at the end of the semester. This approach is designed to give you greater ownership over your learning outcomes, reward your strengths, and encourage reflection on your performance.

Default Grade Weights

- Interactive Checkpoint: 10%
- Labs (approximately 13): 30%
- Exam 1: 15%
- Exam 2: 20%
- Final Exam: 25%

Adjustment Opportunity

Near the end of the semester, after you have received feedback in all categories, you will have the option to redistribute the weights within the ranges below. The total must still add up to 100%, and all categories must remain within their specified limits.

Permitted Weight Ranges:

- Interactive Checkpoint: 5–15%
- Labs: 20–40%
- Exam 1: 10–20%
- Exam 2: 10–30%
- Final Exam: 15–35%

Purpose of This Policy

- Some students excel in practical work, others in exams. This policy allows you to highlight your strongest areas while still maintaining balance.
- Because you will not make your final choice until the end, you will need to engage with all components throughout the semester.
- Choosing your final weights requires you to evaluate your own learning and performance honestly.
- Boundaries prevent extreme weighting changes that would diminish the importance of any one category.

Important Notes

You will make your weight adjustment decision during the final two weeks of the semester. If you do not submit your preferences by the deadline, the default weights will apply. Once submitted, your choice is final and cannot be changed. All weights must fall within the stated ranges and sum to 100%. You will be given a spreadsheet on which you will submit your choice of weights, and we will expect you to submit that spreadsheet over Canvas.

Lab Assignments

- will require programming in C and editing in Vim and/or VS Code using the Unix operating system on our server – hellbender.rnet.missouri.edu
- will require in-person attendance with a fully charged laptop
- will require you to submit your program by the end of your assigned lab session
- will NOT be accepted after the submission deadline stated on the lab assignment document
- will NOT have make-up opportunities

Late labs will NOT be accepted. You must implement your lab program during the assigned lab time. All programming assignments are to be implemented on hellbender.rnet.missouri.edu. The use of this system will be detailed during the class's first lab session. We do not have the resources to assist students who choose to disregard these instructions and use a different system.

We will drop the 2 lowest lab scores as a way to help students who miss a lab due to illness, travel, etc.. Students who participate in a University-sponsored activity that will require them to miss more than 2 labs may request approval to take a proctored remote lab during lab time while they are traveling. However, to take such a proctored remote lab, the student must make arrangements to at the beginning of the semester with the instructor. Students wishing to take a proctored remote lab must use the drop of the 2 lowest lab scores in lieu of labs missed for travel to the University-sponsored activity **before** receiving the opportunity to take a proctored remote lab. For example, a student MAY NOT drop a lab that was missed for illness then drop another lab due to a low score and still expect to then take a proctored remote lab.

Interactive Checkpoint

Interactive Checkpoints will be given during in-person lectures. These will require the use of iClicker software on your mobile device. Login and download the software here: <https://www.iclicker.com/>. Interactive Checkpoint will not be accepted from those not physically in attendance at the lecture.

Bounty Hunting Debugging Sessions

To help you build strong debugging skills (one of the most important abilities in programming) we will begin each Thursday class with a short Bounty Hunting session (sometimes called “Pest Control”). This is a fun, low-stakes activity designed to strengthen your problem-solving skills, get you thinking actively, and make you more confident when working with code.

How it works:

- The first 10–15 minutes of class each Thursday.
- You will work in pairs or trios (you can team up with the nearest classmates) to find and fix 1–2 intentional bugs in a short code snippet.

- The code will be related to recent topics in class (Mostly from the previous lecture on Tuesday). Bugs will be subtle (e.g., logic errors, off-by-one mistakes, variable scope issues) and designed to make you think critically about the code rather than just run it through a tool.
- The code will be shown on slides or printed in class. No standalone files will be provided.

Motivation and Rewards:

- Each group or individual that correctly identifies and fixes the bug(s) earns a “bounty”.
- A Bounty Board will be kept throughout the semester to track earned bounties.
- Students who earn 6 or more bounties will receive a small prize. The prize will be given towards the end of the semester.

Important Notes:

- No course points are awarded for bounties. This is purely for fun, recognition, and skill development.
- These sessions are short and start promptly, arriving on time will ensure you can participate fully.
- The goal is to improve your debugging intuition and teamwork, not just “***get the answer.***”

Why we do this:

- Debugging is a critical skill that can only be improved through consistent practice. These weekly sessions will help you:
 - Quickly identify and diagnose common types of errors.
 - Build strategies for testing and verifying code.
 - Collaborate effectively with peers under time constraints.
 - Start each Thursday’s class with an active, problem-solving mindset.

Midterm and Final Exams

There will be two midterm exams and a final exam. See the course calendar for dates and times. All exams will be held in the regular lecture classroom.

Course Policies

- Class and lab attendance is required!
- NO grade for labs, exams or homework may be changed or disputed more than 1 week after the grade is posted on canvas. I do not accept grade change requests later in the semester. If you see a grade you disagree with then you must submit an email request to me IMMEDIATELY, so I may begin investigating it. After 1 week of posting a grade will not be modified so pay close attention to announcements and canvas where all grades are posted. No late submissions will be accepted for homework assignments.
- No late submissions will be accepted for lab assignments. There will be no makeups of missed lab work.
- Any code that is submitted by a student must be completely original code created independently by that student, unless the code explicitly provides attribution to another source. For example, a student might use code which was presented in class as part of their submission for a homework assignment, but that code must reference the source (in this example, the source would be

something like “CS 1050 Course Notes by Ekincan Ufuktepe”). Failure to reference a source properly will be considered plagiarism and thus an act of academic dishonesty. The use of code not written by the student that constitutes a large percentage of a submission (e.g., using a complete solution from an online source), even if properly attributed, will also be considered plagiarism.

- Any act of academic dishonesty will result in a grade of zero being given to all students involved and a report will be filed with the Provost’s office. You are encouraged to discuss problems and concepts related to the assignments in this class, however you are not allowed to share your answers/solutions/code with any other students. **NO COLLABORATION ALLOWED!**
- Any act of academic dishonesty which calls into question the validity of previously submitted work, may result in a revision of the grades of previously submitted work down to zero points. For example, if a student is found to have cheated on Lab 4, and this leads the instructor to conclude that similar cheating occurred on earlier labs, all labs (including Labs 1, 2, and 3) may be revised to receive zero points for the student in question.
- Make-up exams will only be given in cases where **prior notification** is given to the instructor that **extenuating circumstances** prevent the student from attending the regularly scheduled exam for an **acceptable reason**. It is solely my judgment as to whether I will consider your reason acceptable or not. Please provide notification as early as possible to arrange alternate accommodation. Makeup exams will not be the same as the original exam, which may have a different number of questions, a different format, and, subjectively, a distinct difficulty. I may choose to replace a missed exam score with your Final Exam score (thus over-weighting the Final, which may be a very risky proposition for you).
- No late submissions will be accepted for lab assignments. There will be no make-up for missed lab work.
- Usage of any AI-assisted tools/software/frameworks (e.g. ChatGPT) for any assignments is strictly prohibited. This is a violation of academic integrity, and students who are detected using any AI-assisted tools/software/frameworks will be reported.
- Documentation for Lab instructions and policies will be provided during lab through Canvas.
- None of the class materials can be uploaded to any AI tool, such as ChatGPT. This also involves sharing and distributing course materials outside as well. This is a violation of the TA’s, the instructor’s, and the university’s intellectual property, and legal action will be taken.

Decreasing the Risk of COVID-19 in Classrooms and Labs

Please see updated information throughout the semester at: <https://renewal.missouri.edu/>

Academic Integrity

Academic integrity is fundamental to the activities and principles of a university. All members of the academic community must be confident that each person's work has been responsibly and honorably acquired, developed, and presented. Any effort to gain an advantage not given to all students is dishonest whether the effort is successful or not. The academic community regards breaches of the academic integrity rules as extremely serious matters. Sanctions for such a breach may include academic sanctions from the instructor, including failing the course for any violation, to disciplinary sanctions ranging from probation to expulsion. When in doubt about plagiarism, paraphrasing, quoting, collaboration, or any other form of cheating, consult the course instructor. All exercises and projects are designed and intended to be an individual effort, collaborative implementation or

copying will result in a zero (0) on the exercise for all students involved. Students found to have exhibited academic dishonesty in the execution of programming exercises or projects in this course will be penalized as follows: First occurrence, they will receive a zero on that entire exercise or project. On the second occurrence, the student(s) will receive a zero for that entire course component; e.g., if a student's second instance of academic dishonesty is on a project, they will receive a zero on all projects (completed, graded, and future) in the course for the entire semester.

What happens to me if there is a violation of academic integrity?

When the instructor or the TA detects a violation of academic integrity, it is their responsibility to report (please see [reporting](#) process) it to the academic integrity office. For any violation of academic integrity, the student shall receive a zero for which the violation occurred. A student receiving a zero is not a punishment, it is an academic assessment given by the instructor or the TA, due to the student not showing the required performance and effort to complete the given assignment/lab/exams etc.. The violation will be assessed, and the punishment will be determined by the academic integrity office.

What are the types of academic integrity violation?

Academic integrity is fundamental to the activities and principles of a university. All members of the academic community must be confident that each person's work has been responsibly and honorably acquired, developed, and presented. Any effort to gain an advantage not given to all students is dishonest whether or not the effort is successful. The academic community regards breaches of the academic integrity rules as extremely serious matters. Sanctions for such a breach may include academic sanctions from the instructor, including failing the course for any violation, to disciplinary sanctions ranging from probation to expulsion. When in doubt about plagiarism, paraphrasing, quoting, collaboration, or any other form of cheating, consult the course instructor. According to the standard of conduct:

The term **cheating** includes but is not limited to: (i) use of any unauthorized assistance in taking quizzes, tests, or examinations; (ii) dependence upon the aid of sources beyond those authorized by the instructor in writing papers, preparing reports, solving problems, or carrying out other assignments; (iii) acquisition or possession without permission of tests or other academic material belonging to a member of the University faculty or staff; or (iv) knowingly providing any unauthorized assistance to another student on quizzes, tests, or examinations.

The term **plagiarism** includes, but is not limited to: (i) use by paraphrase or direct quotation of the published or unpublished work of another person without fully and properly crediting the author with footnotes, citations or bibliographical reference; (ii) unacknowledged use of materials prepared by another person or agency engaged in the selling of term papers or other academic materials; or (iii) unacknowledged use of original work/material that has been produced through collaboration with others without release in writing from collaborators.

The term **unauthorized use of artificially generated content**, includes, but is not limited to (i) use of artificial intelligence tools or other tools that generate artificial content in taking quizzes, tests, examinations, or other assessments without permission from the instructor; (ii) submitting work for evaluation as one's own that was produced in material or substantial part through use of artificial intelligence tools or other tools that generate artificial content without permission from the

instructor; (iii) using artificial intelligence tools or other tools that generate artificial content in a manner contrary to instructions from the instructor; or (iv) using artificial intelligence tools or other tools that generate artificial content in a manner that violates any other provision of these rules concerning academic dishonesty. Use of commonly available tools such as spelling or grammar checking software or features of software that propose anticipated words or phrases while text is being written will not be considered unauthorized use of artificially generated content unless such use is contrary to instructions from the instructor.

The term **sabotage** includes, but is not limited to, the unauthorized interference with, modification of, or destruction of the work or intellectual property of another member of the University community.

For more information about academic integrity, please see the link below:

[Office of Academic Integrity \(missouri.edu\)](https://academicintegrity.missouri.edu)

ADA Notice

Students with Disabilities: If you anticipate barriers related to the format or requirements of this course, if you have emergency medical information to share with me, or if you need to make arrangements in case the building must be evacuated, please let me know as soon as possible. If disability related accommodations are necessary (for example, a note taker, extended time on exams, captioning), please register with the Disability Center (<http://disabilitycenter.missouri.edu>), S5 Memorial Union, 573-882-4696, and then notify me of your eligibility for reasonable accommodations. For other MU resources for persons with disabilities, click on "Disability Resources" on the MU homepage.

Intellectual Pluralism

The University community welcomes intellectual diversity and respects student rights. Students who have questions or concerns regarding the atmosphere in this class (including respect for diverse opinions) may contact the Departmental Chair or Divisional Director; the Director of the Office of Students Rights and Responsibilities (<http://osrr.missouri.edu/>); or the MU Equity Office (<http://equity.missouri.edu/>), or by email at equity@missouri.edu. All students will have the opportunity to submit an anonymous evaluation of the instructor(s) at the end of the course.

Recording Course Activities

University of Missouri System Executive Order No. 38 lays out principles regarding the sanctity of classroom discussions at the university. The policy is described fully in Section 200.015 of the Collected Rules and Regulations. In this class, students may make audio or video recordings of course activity unless specifically prohibited by the faculty member. However, the redistribution of audio or video recordings of statements or comments from the course to individuals who are not students in the course is prohibited without the express permission of the faculty member and of any students who are recorded. Students found to have violated this policy are subject to discipline in accordance with provisions of Section 200.020 of the Collected Rules and Regulations of the University of Missouri pertaining to student conduct matters.

Religious Observances

University of Missouri does not restrict students free exercise of religion, unless 1) the restriction is in the form of a rule of general applicability, and does not discriminate against religion, or among religions; and 2) it can be demonstrated that the application of the restriction is essential to furthering a compelling university interest, and is not unduly restrictive considering the relevant circumstance. The policy of the University attempts to strike a reasonable balance between accommodating the religious practice of students and meeting academic needs and standards. Students may not be penalized for absence from class to participate in religious observances; however, students should notify instructors as soon as possible about such absences by submitting the form available at <https://diversity.missouri.edu/wp-content/uploads/2018/07/university-of-missouri-student-request-foraccommodation-of.pdf>. If such an absence coincides with the exam/assignment, make-up exam policy will be regarded.

Mental Health and Stress Management

The University of Missouri is committed to supporting student well-being through an integrated network of care, with a wide range of services to help students succeed. The MU Counseling Center offers professional mental health care, and can help you find the best approach to treatment based on your needs. Call to make an appointment at 573-882-6601. Any student in crisis may call or go to the MU Counseling Center between 8:00 – 5:00 M-F. After hours phone support is available at 573-882- 6601.

Visit our website at <https://wellbeing.missouri.edu> to take an online mental health screening, find out about workshops and resources that can help you thrive, or learn how to support a friend.

Download Sanvello, a phone app that teaches skills and strategies to help you maintain good mental health. Log in with your Mizzou e-mail to unlock all the tools available through Sanvello at no cost to you.

Sexual Harassment and Misconduct:

For complete information, please refer to:

https://www.umsystem.edu/ums/rules/collected_rules/equal_employment_educational_opportunity/ch600/600.020_sex_discrimination_sexual_harassment_and_sexual_misconduct .